

Electricity-Visualisation Milestone 2

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Tools and courses needed

To build our visualizations, we will use HTML, CSS, and JavaScript for the website's basic structure, alongside Scrollama.js to handle the scrollytelling mechanics. We will rely heavily on D3.js to create the interactive price line charts and the fluid flow maps, while using Python and Pandas offline to clean and process our dataset. To successfully implement these tools, we will need to apply concepts from several course lectures, specifically the introduction to D3 and SVG, transitions and animations in D3, geospatial data mapping, and the lecture on interaction and narrative scrollytelling.

While working on the skeleton, we have implemented some samples of the visual designs we plan on creating for the finished product and we have understood the basic mechanics of the tools cited above but the full project will require more studying of the documentation to understand the full potential of these libraries.

Detailed Project Architecture: From Macro Conflicts to Micro Impacts

Our project aims to show how geopolitical conflicts affect global electricity supply, demand, and prices. We are building an interactive website to help people understand how these distant events impact their daily lives and wallets.

(sketch 1: Conflict Choice)

The experience begins with the user selecting between the Ukrainian conflict and the Iranian conflict. This choice determines the rest of the storyline: both the map and the subsequent slides dynamically adapt to reflect the selected conflict. We believe this branching design improves the website's fluidity while keeping the experience intuitive, as each user follows a shorter, more focused timeline rather than navigating through all scenarios at once.

We will design this slide in the following way:

Each conflict will be represented by one flag in the shape of the country where the conflict happened. These flags are buttons that the user should click on to make the choice. We will need to study how to make it clear to the user that he is supposed to click to continue. Our initial thought is to include a little text that appears after 5 seconds reading "Select a conflict to continue".

Once the button is clicked the website moves to the next page sliding down.

(Sketch 2: The Interactive Map and Time Slider)

This slide is made up of an interactive map covering the whole website's screen. This map is a visual representation of evolution of energy prices in Europe. We plan on making this map interactive by adding a slider element that controls the timeline of the map. As the user progresses through the slider, the colours on the map change reflecting prices evolution across time.

The choice of colours is still to be determined but we will probably work with soft tones of blue to red.

Once the user is ready to move on, he will need to slide down with his mouse or trackpad. To indicate that this movement is expected of the user, we will add a small appearing arrow at the bottom of the screen.

(Sketch 3: The Shop Shelf with Everyday Items)

Finally the user arrives in a visual shop where he will see grocery items stacked on a shelf. The user will be able to hover over items. On hover, a text box will appear next to the mouse. The text box will contain information about the selected product (name, price before conflict, price after).

The goal of this window is to help the user understand directly how the price products he consumes everyday are affected by conflicts and time.

We believe that this slide will be the hardest to design as it requires seamless interactivity, precise data integration, and a clear yet intuitive visual layout to effectively link product prices with geopolitical events over time.

Finally, if time permits, we may enhance the project with extra features like a conflict receipt, a historical news ticker, animated energy flows.

New ideas :

To enhance our visualization, we have a few extra ideas that can be safely dropped if needed. We could add a conflict receipt: a virtual shopping cart that calculates exactly how much extra money the crisis is costing the user.

The user would be able to select some items from the shelf to build a shopping cart. When is ready the user will be asked to click on a button. On action, a receipt would printed out and shown. On the receipt we will find the price of the product before and after the conflict, and finally a "before and after" total value of the whole shopping cart.

Although this would be hard to design we think that it would be a fun addition to the project and increase user interactivity.

Also we might include a scrolling news ticker showing real time news headlines that match the selected date below the interactive map. This would allow the user to understand how the conflict evolved in time and directly see its effect on electricity prices.

Finally, we could display animated energy flows using moving dots to represent electricity exchanges between countries across Europe. These flows would help illustrate how energy is redistributed geographically in response to shocks and imbalances caused by the conflicts.

To make this visualization more intuitive, we plan to encode price dynamics directly into the animation: flows would gradually turn blue when electricity prices decrease and red when prices increase. This colour transition would provide an immediate visual indication linking cross-border energy movements to market conditions.

However, implementing this feature would require access to detailed cross-border electricity flow data at a sufficiently high temporal resolution (at least monthly or quarterly). As we do not currently have access to such data, we would need to find an appropriate data source.